CONVERGENCE OF THE KUNNETH SPECTRAL SEQUENCES FOR A BOUNDED COMPLEX

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

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Allowed standard prerequisites (editorial).

- Spectral sequence of a filtered complex and the convergence criterion for a bounded filtration. used throughout the proof, lines 15543-15643. Supplies the object whose convergence is asserted and the criterion checked at the end.
- The Kunneth spectral sequences for the tensor product and for the Hom complex. constructed in the same section, source lines 15300-15516, before the counted statement. Provides the two spectral sequences the statement refers to.
- Identification of the graded pieces of a filtration with Tor and Ext terms. used at proof lines 15560-15620. Supplies the second page in both cases.

Source reference. Original lemma 0H7W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 21471–21471. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 012W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 1414–1414. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0BK5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14548–14548. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Lemma 0.2. *Under the assumptions above, if in addition*

(1) *the filtration on $K^\bullet$ is finite and the filtration on $L^\bullet$ is finite, or*
(2) *more generally the following are true*
   (a) *$F^iK^\bullet$ is acyclic for $i \gg 0$,*
   (b) *$F^iK^\bullet \to K^\bullet$ is a quasi-isomorphism for $i \ll 0$,*
   (c) *$F^jL^\bullet$ is acyclic for $j \gg 0$, and*
   (d) *$F^jL^\bullet \to L^\bullet$ is a quasi-isomorphism for $j \ll 0$.*

Then the spectral sequence is bounded, the associated filtration on each $H^n(T^\bullet) = H^n(K^\bullet \otimes^{\mathbf{L}}_R L^\bullet)$ is finite and we have convergence

$$E_1^{p,q} = \bigoplus\nolimits_{i+j=p} H^{p+q}(gr^iK^\bullet \otimes^{\mathbf{L}}_R gr^jL^\bullet) \Rightarrow H^n(K^\bullet \otimes^{\mathbf{L}}_R L^\bullet)$$

Proof. In case (1) the filtration on $T^\bullet$ is finite and the lemma follows immediately from Homology, Lemma 012W. In case (2) choose $a < b$ such that $F^iK^\bullet$ and $F^jL^\bullet$ are acyclic for $i, j > b$ and $F^iK^\bullet \to K^\bullet$ and $F^jL^\bullet \to L^\bullet$ are quasi-isomorphisms for $i, j < a$. We claim that in this case the complex $F^nT^\bullet$ is acyclic for $n > 2b$ and that $F^nT^\bullet \to T^\bullet$ is a quasi-isomorphism for $n < 2a-1$. Since the claim shows that Homology, Lemma 0BK5 applies we see that our lemma is true.

Proof of the claim. Given n and integers $i_1 \leq i_2$ consider

$$S^{n,i_1,i_2} = \mathrm{Im}\left(\bigoplus\nolimits_{i_1 \leq i \leq i_2} \mathrm{Tot}(F^iK^\bullet \otimes_R F^{n-i}L^\bullet) \to \mathrm{Tot}(K^\bullet \otimes_R L^\bullet)\right)$$

viewed as a subcomplex of $\mathrm{Tot}(K^\bullet \otimes_R L^\bullet)$. Observe that $F^nT^\bullet = \mathrm{colim}\, S^{n,i_1,i_2}$ is a filtered colimit. Thus to show the claim it suffices to show that S^{n,i_1,i_2} is acyclic for $n > 2b$ and that $S^{n,i_1,i_2} \to \mathrm{Tot}(K^\bullet \otimes L^\bullet)$ is a quasi-isomorphism for $n < 2a-1$ and a cofinal set of choices of pairs i_1, i_2.

If $i_1 = i_2 = i$ and $n > 2b$, then

$$S^{n,i,i} = \mathrm{Tot}(F^iK^\bullet \otimes_R F^{n-i}L^\bullet)$$

is acyclic: either $i > b$ and this complex is acyclic by our assumption that $F^{n-i}L^\bullet$ is K-flat, or $i \leq b$ and then $n - i \geq n - b > b$ and the same holds. Using our flatness assumptions the reader shows that there is a short exact sequence

$$0 \to S^{n+1,i_2+1,i_2+1} \to S^{n,i_1,i_2} \oplus S^{n,i_2+1,i_2+1} \to S^{n,i_1,i_2+1} \to 0$$

of complexes. Thus we see by induction on $i_2 - i_1$ that all of the complexes S^{n,i_1,i_2} for $n > 2b$ are acyclic.

Assume $n < 2a-1$. Observe that

$$S^{n,a-1,a-1} = \mathrm{Tot}(F^{a-1}K^\bullet \otimes_R F^{n-a+1}L^\bullet) \to \mathrm{Tot}(K^\bullet \otimes_R L^\bullet)$$

is a quasi-isomorphism because both $a - 1 < a$ and $n - a + 1 < a$ and all our complexes are K-flat hence both sides compute the same object of $D(R)$. Observe that for $i_2 \geq a - 1$ and $n < 2a - 1$ the map

$$S^{n+1,i_2+1,i_2+1} \to S^{n,i_2+1,i_2+1}$$

is a quasi-isomorphism because left and right hand side are quasi-isomorphic to $\mathrm{Tot}(F^{i_2+1}K^\bullet \otimes L^\bullet)$. Thus using the short exact sequence of complexes in the previous paragraph we find that $S^{n,a-1,a-1} \to S^{n,a-1,t}$ is a quasi-isomorphism for all $t \geq a - 1$. On the other hand, there are similarly short exact sequences

$$0 \to S^{n+1,i_1,i_1} \to S^{n,i_1-1,i_1-1} \oplus S^{n,i_1,i_2} \to S^{n,i_1-1,i_2} \to 0$$

of complexes. Similarly to the above, we note that for $i_1 \leq a - 1$ the map

$$S^{n+1,i_1,i_1} \to S^{n,i_1-1,i_1-1}$$

is a quasi-isomorphism because left and right hand side are quasi-isomorphic to $\mathrm{Tot}(K^\bullet \otimes F^{n-i_1+1}L^\bullet)$. Thus we conclude that $S^{n,s,t} \to S^{n,a-1,t}$ is a quasi-isomorphism for all $s \leq a-1$. Combined we find that $S^{n,s,t}$ for $s \leq a-1$ and $t \geq a-1$ maps quasi-isomorphically to $\mathrm{Tot}(K^\bullet \otimes_R L^\bullet)$. This finishes the proof of the lemma. $\square$